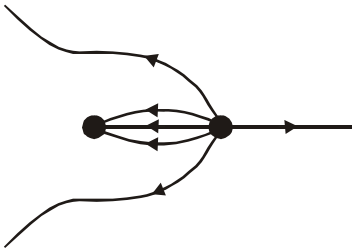
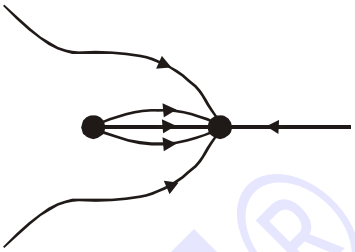
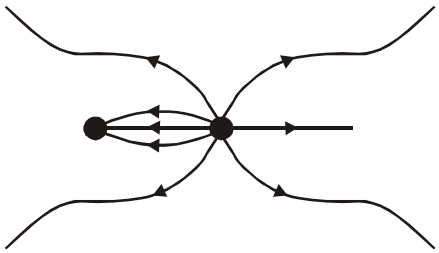
SECTION-I

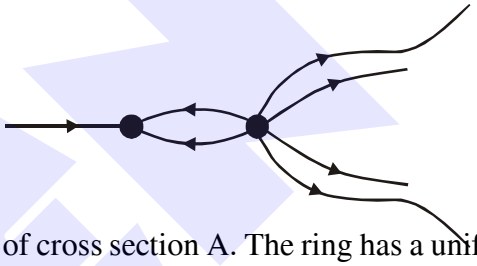
Single Correct Answer Type **10 Q. [3 M (–1)]**

1. Two point charges of –3 C and + 6 C are placed, choose most appropiate field line diagram :-

(A) 

(B) 

(C) 

(D) 

2. A ring of radius R is made out of a thin wire of area of cross section A. The ring has a uniform charge Q distributed on it. A charge q_0 is placed at the centre of the ring. If Y is the young's modulus for the material of the ring and ΔR is the change in the radius of the ring then,

(A) $\Delta R = \frac{q_0 Q}{4\pi\varepsilon_0 RAY}$ (B) $\Delta R = \frac{q_0 Q}{4\pi^2\varepsilon_0 RAY}$ (C) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0^2 RAY}$ (D) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0 RAY}$

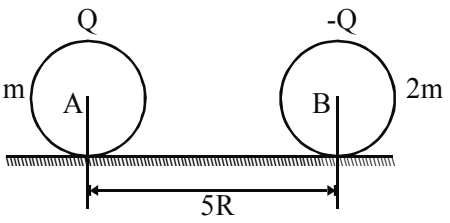
3. The electrostatic potential V at a point on the circumference of a thin non–conducting disk of radius r and uniform charge density σ is given by equation $V = 4\sigma r$. Which of the following expression correctly represents electrostatic energy stored in the electric field of a similar charged disk of radius R?

(A) $U = \frac{8}{3}\pi\sigma^2 R^3$ (B) $U = \frac{4}{3}\pi\sigma^2 R^3$ (C) $U = \frac{2}{3}\pi\sigma^2 R^3$ (D) None of these

4. Two smooth spherical non conducting shells each of radius R having uniformly distributed charge Q and –Q on their surfaces are released on a smooth non-conducting surface when the distance between their centres is 5R. The mass of A is m and that of B is 2m. The speed of A just before A and B collide is [Neglect gravitational interaction] $\left(K = \frac{1}{4\pi \in_0}\right)$

(A) $\sqrt{\frac{2KQ^2}{5mR}}$ (B) $\sqrt{\frac{4KQ^2}{5mR}}$

(C) $\sqrt{\frac{8KQ^2}{5mR}}$ (D) $\sqrt{\frac{16KQ^2}{5mR}}$



5. Charge q and $-\frac{cq}{b}$ are placed at (b,0) and $\left(\frac{c^2}{b}, 0\right)$. The points of zero potential make a locus in the x–y plane. What is its shape ?

(A) Circle (B) Parabola (C) Hyperbola (D) None of these

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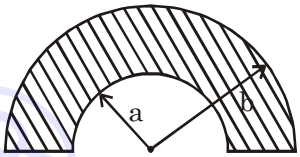
6. Two small spheres have mass m_1 and m_2 and hanging from massless insulating threads of lengths ℓ_1 and ℓ_2. Two sphere carry charges q_1 and q_2 respectively. The spheres hang such that they are on the same horizontal level and the threads are inclined to the vertical at angle θ_1 and θ_2 respectively. Which of the following condition is true if $m_1 = m_2$.

(A) $\theta_1 = \theta_2$ (B) $q_1 = q_2$ (C) $\frac{\ell_1}{\tan\theta_1} = \frac{\ell_2}{\tan\theta_2}$ (D) $\frac{q_1}{\tan\theta_1} = \frac{q_2}{\tan\theta_2}$

7. A non conducting semicircular disc (as shown in figure) has a uniform surface charge density σ. The electric field intensity at the centre of the disc

(A) $\frac{\sigma(b-a)}{4\pi\varepsilon_0}$ (B) $\frac{\sigma(b-a)}{2\pi\varepsilon_0}$

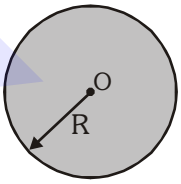
(C) $\frac{\sigma\ln(b-a)}{2\pi\varepsilon_0(b-a)}$ (D) $\frac{\sigma}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right)$



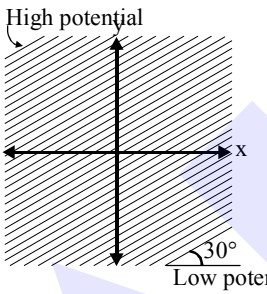
8. Volume charge density of a non–conducting sphere is varying as $\rho = \rho_0 r$ where ρ_0 is a constant and r is the distance from centre of sphere.If potential at r=2R is zero then potential at infinity will be

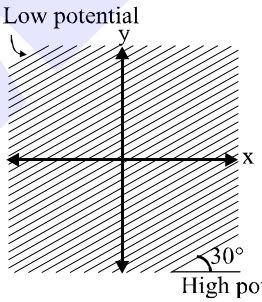
(A) $\frac{\rho_0 R^3}{8\epsilon_0}$ (B) $-\frac{\rho_0 R^3}{8\epsilon_0}$

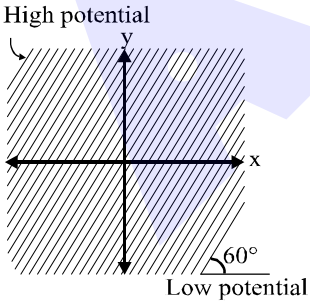
(C) $-\frac{\rho_0 R^3}{4\epsilon_0}$ (D) Data insufficient

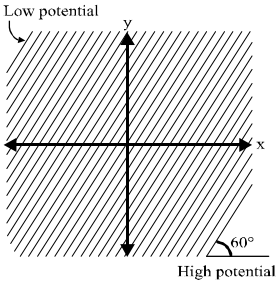


9. The electric field intensity at all points in space is given by $\vec{E} = \sqrt{3}\hat{i} - \hat{j}$ volts/metre. The nature of equipotential lines in x-y plane is given by :-

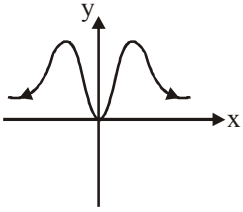
(A)


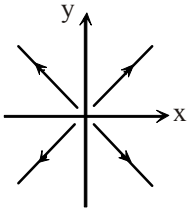
(B)


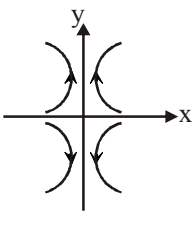
(C)


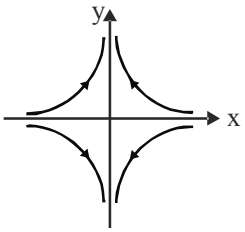
(D)


10. In a certain region of space, the potential field depends on x and y coordinates as $V = (x^2 - y^2)$. The corresponding electric field lines in x-y plane are correctly represented by :

(A)


(B)


(C)


(D)


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Multiple Correct Answer Type 4 Q. [4 M (–1)]

11. Electric field in a region is given as $\vec{E} = (10 - 5x)\hat{i}$. A charge particle of mass 5kg and charge Q (= 1C) is situated at origin and free to move in given electric field. Then choose the correct options (Neglect gravity) :-
(A) Motion of charge particle is Oscillatory
(B) Maximum displacement of charge particle from origin is 4 SI units
(C) Maximum velocity gain by charge particle is 2 SI units
(D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units

12. Four identical charges are placed at the points (1, 0, 0), (0, 1, 0), (–1, 0, 0) and (0, –1, 0) :-
(A) The potential at the origin is zero.
(B) The field at the origin is zero.
(C) The potential at all points on the z-axis, other than the origin, is zero.
(D) The field at all points on the z-axis, other than the origin acts along the z-axis.

13. A stationary electric dipole $\vec{\phi} = p\hat{k}$ is situated at origin. A positive charge q, mass m executes circular motion of radius s at constant speed in the field of the dipole. Characterize the plane of orbit is :-

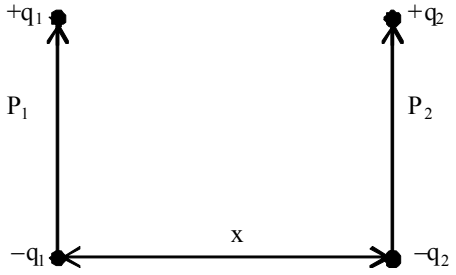
(A) The speed of charge particle is $v = \frac{1}{s}\sqrt{\frac{2qp}{\sqrt{3}\pi \epsilon_0 m}}$

(B) The speed of charge particle is $v = \frac{1}{s}\sqrt{\frac{qp}{3\sqrt{3}\pi \epsilon_0 m}}$

(C) Angular momentum of charge is $L = \sqrt{\frac{2qpm}{\sqrt{3}\pi \epsilon_0}}$

(D) Angular momentum of charge is $L = \sqrt{\frac{qpm}{3\sqrt{3}\pi \epsilon_0}}$

14. Figure shows two dipole moments parallel to each other and placed at a distance x apart x is very large, then :
(A) they will repel each other
(B) they will attract each other



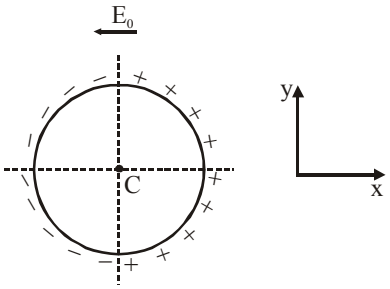
(C) force of interaction is of magnitude of $\frac{3P_1P_2}{4\pi\varepsilon_0 x^4}$

(D) force of interaction is of magnitude of $\frac{6P_1P_2}{4\pi\varepsilon_0 x^4}$

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question No. 15 and 16

A uniform ring of mass m and radius R can rotate freely about an axis passing through centre C and perpendicular to plane of paper. Half of ring is positively charge and other half is negatively charge. Uniform electric field E_0 is switched on along –ve x-axis (axis are shown in figure) [Magnitude of charge density λ]



15. The dipole moment of ring is :-
(A) $2\lambda R^2$ (B) $4\lambda R^2$
(C) $2\pi\lambda R^2$ (D) $4\pi\lambda R^2$

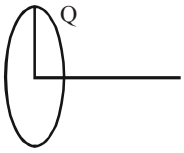
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16. If ring is slightly disturb from given position, find the angular speed of ring when it rotate by $\pi/2$.

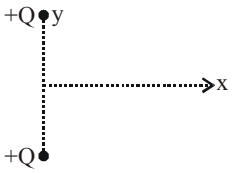
(A) $2\sqrt{\frac{\lambda E_0}{m}}$ (B) $\sqrt{\frac{\lambda E_0}{m}}$ (C) $\sqrt{\frac{8\lambda E_0}{m}}$ (D) None

Matching List Type (4 × 4) 1Q.[3 M (–1)]

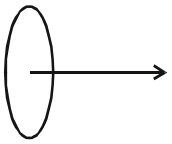
17. **List-I** **List-II**

(P) 

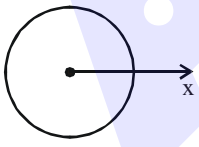
Uniformly positively charged ring in yz plane with axis along x axis

(Q) 

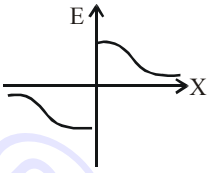
Two identical positively charged particle placed on y axis symmetrical to x axis

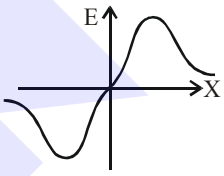
(R) 

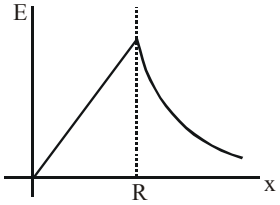
Uniformly positively charged non conducting disc in yz plane with axis along x axis.

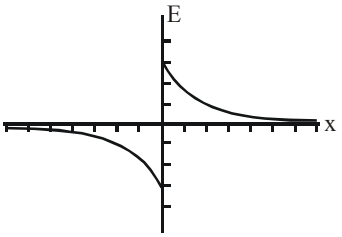
(S) 

Uniformly positively charged (charge distribut throughout its volume) non conducting sphere with diameter axis along x-axis.

(1) 

(2) 

(3) 

(4) 

Code :-

	P	Q	R	S
(A)	1	2	3	4
(B)	2	2	4	3
(C)	3	2	1	4
(D)	2	1	3	4

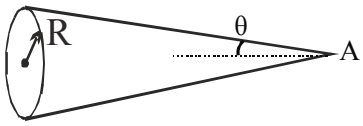
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SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

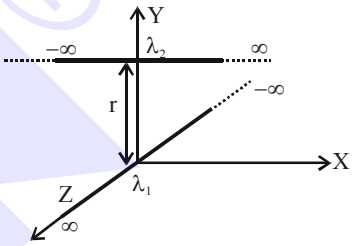
18. A cone made of insulating material has a total charge Q = 3mC spread uniformly over its sloping surface. Calculate the energy (in Joule) required to take a charge q = 5μC from infinity to apex A of cone. (Given : half cone angle θ = 37°, R = 2m)



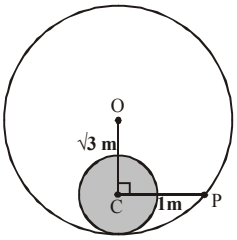
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **9 Q. [4 M (0)]**

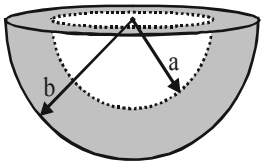
19. AB and CD are uniform line charges of infinite length having charge density λ_1 and λ_2 and lying parallel to the Z–axis and X–axis respectively. The force between them depends on the perpendicular distance between them given by 'r' as shown in figure. The value of $F = \frac{\lambda_1\lambda_2}{2\varepsilon_0 r^n}$, then write the value of n in OMR sheet.



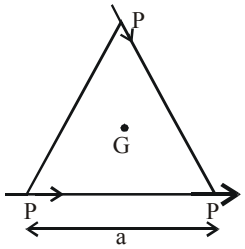
20. A uniformly charged sphere is placed inside a charged hollow sphere as shown in figure. O is the centre of hollow sphere and C is the centre of solid sphere. The magnitude of charge on both the spheres is $4\left(\sqrt{\frac{3}{7}}\right)\mu C$. The electric field at point 'P' which lie just outside the hollow sphere is given by 3α kN/C . Find the value of α.



21. A uniformly charged hemisphere of radius b and charge density ρ has a hemispherical cavity of radius $a\left(a = \frac{b}{2}\right)$ cut from its centre. If the potential at the centre of the cavity is $\frac{n\rho b^2}{16\epsilon_0}$ then n = ?

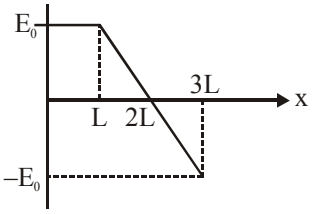


22. Three small dipoles of dipole moment P each are placed at the three vertices of an equilateral triangle of side a as shown. If total potential at the centroid G is given by $V = \frac{n\times 3\sqrt{3}kp}{2a^2}$. Find $n\left(k = \frac{1}{4\pi\epsilon_0}\right)$.



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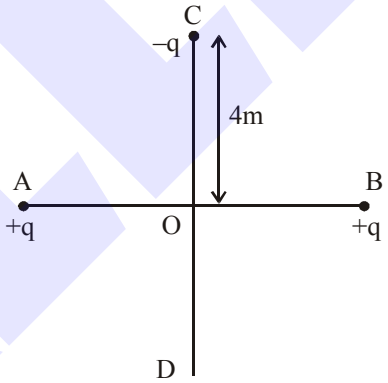
23. Consider the E versus x graph. What is the minimum velocity that should be given to a point charge –Q of mass M at x = 3L so that it can reach the origin? Take Q = 1 mC, mass of the charge as 1g, E_0 = 5 N/C and L = 5m.



24. A small point charge (+q = 1μC, m = 3 kg) is started from a very long distance towards a fixed point charge (+Q = 1 mC). The initial line of motion of q is at a ⊥ distance of d = 1m from the line passing from the centre of Q, (as shown in figure). Take V_0 = 2 m/s. Determine the least distance between the two point charges (in m)



25. Two fixed, equal positive charges, each +q are located at point A & B separated by a distance of 6 m. A particle of mass m having equal and opposite charge –q moves towards them along the perpendicular bisector line COD where O is the centre of line joining A and B. If –q charge is released from rest from point C, then speed of charge –q at O is given by $v = q\sqrt{\frac{x}{15\pi \in_0 m}}$ find value of x. (Neglect gravity)



26. The electric potential in a region is given by $V(x, y, z) = ax^2 + ay^2 + abz^2$. 'a' is a positive constant of appropriate dimensions and b, a positive constant such that V is volts when x, y, z are in m. Let b = 2. The work done by the electric field when a point charge +4μC moves from the point (0, 0, 0.1m) to the origin is 50 μJ. The radius of the circle of the equipotential curve corresponding to V = 6250 volts and $z = \sqrt{2}$ m is α m. Fill α^2 in OMR sheet.

27. The electric field strength depends only on the x, y and z coordinates according to the law $E = \frac{a(x\hat{i} + y\hat{j} + z\hat{k})}{\left(x^2 + y^2 + z^2\right)^{3/2}}$, where a = 122.5 SI unit and is a constant. Find the potential difference (in volt) between (3, 2, 6) and (0, 3, 4).

For More Material Join: @JEEAdvanced_2024

Subjective Type **2 Q. [4 M (0)]**

28. Electric dipole of moment $P = P\hat{i}$ is kept at a point (x,y) in an electric field, $E = 4xy^2\hat{i} + 4x^2y\hat{j}$. Find the forces on the dipole.

29. Four point charges $+8\mu C$, $-1\mu C$, $-1\mu C$ and $+8\mu C$, are fixed at the points, $-\sqrt{\frac{27}{2}}$ m, $-\sqrt{\frac{3}{2}}$ m, $+\sqrt{\frac{3}{2}}$ m and $+\sqrt{\frac{27}{2}}$ m respectively on the y-axis. A particle of mass 6×10^{-4} kg and of charge + 0.1 μC moves along the $-x$ direction. Its speed at $x = +\infty$ is v_0. Find the least value of v_0 for which the particle will cross the origin . Find also the kinetic energy of the particle at the origin. Assume that space is gravity free. (Given : $1/(4\pi\varepsilon_0) = 9 \times 10^9$ Nm^2/C^2)

GR # ELECTRIC FIELD & POTENTIAL **ANSWER KEY**

SECTION-I

Single Correct Answer Type **10 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (D)** **3. Ans. (A)** **4. Ans. (A)** **5. Ans. (A)** **6. Ans. (A)**
7. Ans. (D) **8. Ans. (B)** **9. Ans. (C)** **10. Ans. (D)**

Multiple Correct Answer Type **4 Q. [4 M (–1)]**

11. Ans. (A,B,C,D) **12. Ans. (B,D)** **13. Ans. (B, D)** **14. Ans. (A,C)**

Linked Comprehension Type **(1 Para × 2Q.)** **[3 M (-1)]**
(Single Correct Answer Type)

15. Ans. (B) **16. Ans. (C)**

Matching List Type (4 × 4) **1Q.[3 M (–1)]**

17. Ans. (B)

SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**
(upto second decimal place)

18. Ans. 81

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **9 Q. [4 M (0)]**

19. Ans. 0 **20. Ans. 9** **21. Ans. 3** **22. Ans. 1** **23. Ans. 5** **24. Ans. 2**
25. Ans. 2 **26. Ans. 6** **27. Ans. 7**

Subjective Type **2 Q. [4 M (0)]**

28. Ans. $4py\ (y^2 + 4x^2)^{1/2}$

29. Ans. $v_0 = 3$ m/s ; K.E. at the origin $= \left(27 - 10\sqrt{6}\right) \times 10^{-4}$ J approx. 2.5×10^{-4} J

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